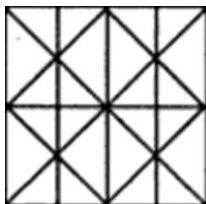


it. ANALYTICAL REASONING

The chapter on Analytical Reasoning involves the problems relating to the counting of geometrical figures in a given complex figure. The systematic method for determining the number of any particular type of figure by the analysis of the complex figure would be clear from the examples that follow.

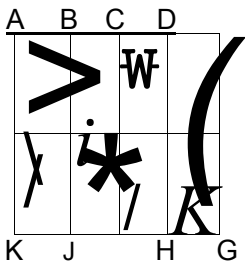
Ex. 1 ; What is the number of straight lines In the following figure?



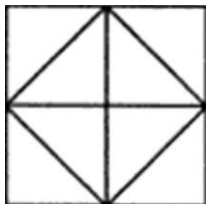
- (a) 11
- (b) 14
- (3) 16
- (d) 17

Sol. The figure is labelled as shown.

Clearly, there are 3 horizontal lines namely AE, LF and KG.
There are 5 vertical lines : AK, BJ, CI, DH and EG.
There are 6 slanting lines : LC, KE, IF, LI, AG and CF.
Thus, there are $3 + 5 + 6 = 14$ straight lines in the figure.
Hence, the answer is (to).



Ex. 2 ; Count the number of triangles In the following figure.



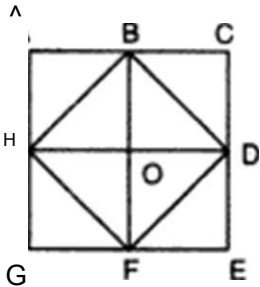
- (a) 8
- (to) 10
- (3) 12
- (d) 14

Sol. We first label the entire figure as shown.

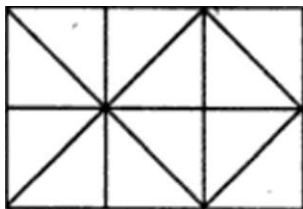
Count the number of simplest triangles. These are ABH, BHO, BCD, BOD, DEF, DFO, FGH and FHO. Thus, there are 8 such triangles.
Next count the number of triangles which are composed of two components each. Such triangles are HBD, BDF, DFH and FHB. Thus, there are 4 such triangles.

Triangles with more than two components do not exist in the given figure.

.% The total number of triangles in the given figure » $8 + 4 = 12$
Thus, (c) is the answer.



Ex. 3 ; How many squares does the figure have?



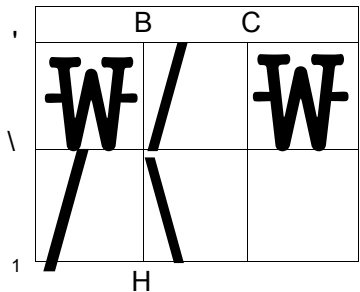
- (a) 6
- (to) 7
- (0) 9
- (d) 10

Sol. The figure may be labelled as shown :

The squares composed of two components each, are ABKJ, BCLK, CDEL, LEFG, KLGH and JKHI. Thus, there are 6 such squares.

Only one square, KCEG is composed of four components
Two squares namely, ACGI and BDFH are composed of eight components each. Thus, there are 2 such squares.

There are $6 \cdot 1 \cdot 2 = 9$ squares in the given figure.
Hence, (c) is the answer.



Ex. 4 : How many parallelograms are there in the figure below ?

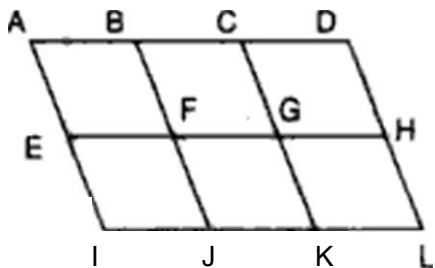


- (a) 14
- to 15
- (d) 18

Sol. We can label the figure as shown.

The simplest || gms are ABFE, BCGF, CDHG, EFJI, FGKJ AND GHKL. These are 6 in number.

The || gms composed of two components each, are ACEG, BDHF, EGKI, FHU, ABJI, BCKJ, and CDLK. Thus, there are 7 such || gms.



The || gms composed of four components each, are ACKI and BDLJ i.e. 2 in number.
There is only one || gm composed of six components, namely, ADLI.
Thus, there are $6+7+2+1=16$ parallelograms in the figure.
Hence, (c) is the answer.

Ex 5. What is the number of rectangles in the following figure?

- (a) 6
- to 8
- (b) 7
- to 9

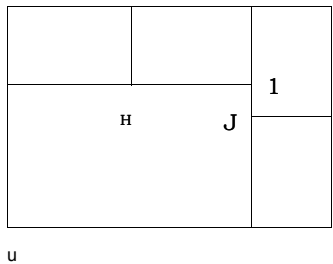
Sol. The figure is labelled as shown :

Simplest rectangles are AEHG, EFIH, FBKJ, JKCL and GILD. i.e. there are 5 such rectangles

The rectangles composed of two components each are AFIG and FBCL. Thus, there are 2 such rectangles.

Only one rectangle, namely AFLD is composed of 3 components and only one rectangle, namely ABCD is composed of 5 components

Thus, there are $5 + 2 \cdot 1 + 1 = 9$ rectangles in the figure
Hence, (d) is the answer.



Ex. 6 ; Determine the number of pentagons in the following figure



- (a) 5
- to 6
- (c) 8
- to 9

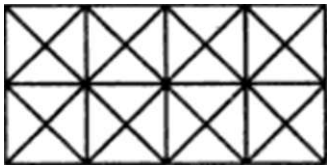
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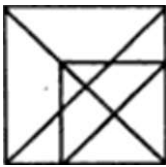
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32. Count the number of squares in the figure given below



- (a) 11
- (b) 21
- (c) 24
- (d) 26

33. How many triangles are there in the figure given below?



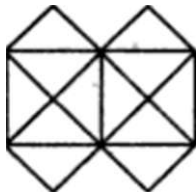
- (a) 16
- (b) 18
- (c) 19
- (d) 20

34. How many squares does the following figure have?



- (a) 22
- (b) 20
- (c) 18
- (d) 16

Directions ; Study the following figure and answer questions 35 to 37.



35. What is the minimum number of straight lines that is needed to construct the figure ?

- (a) 11
- (b) 13
- (c) 15
- (d) 21

36. Count the number of triangles in the figure.

- (a) 12
- (b) 16
- (c) 20
- (d) 24

37. How many squares does the figure contain?

- (a) 5
- (b) 6
- (c) 7
- (d) 8

38. How many squares are there in the following figure?

- (a) 16
- (b) 17
- (c) 25
- (d) 27

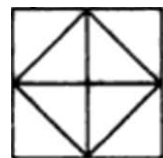
39. Count the number of triangles and parallelograms in the figure given below



- (a) 16. 22
- (b) 18. 16
- (c) 14. 20
- (d) 15. 21

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46. Count the number of pentagons in the following figure

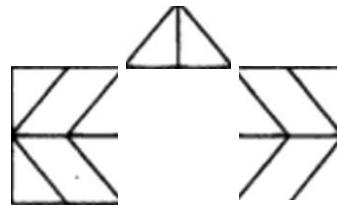


- (a) 16

(e) 12
- (b) 14

(d) 10

49. Determine the number of rectangles and hexagons in the following figure



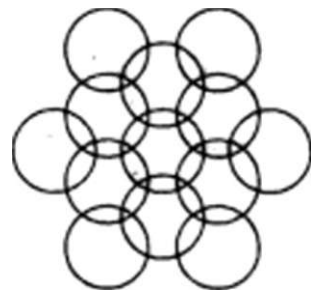
- (a) 8 rectangles, 3 hexagons

(b) 15 rectangles, 3 hexagons

(c) 24 rectangles, 5 hexagons

(d) 30 rectangles, 5 hexagons

50. How many circles are there in the figure given below?



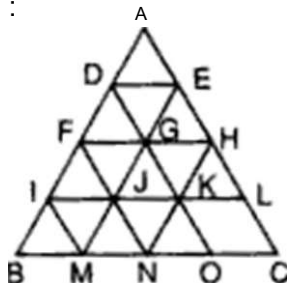
- (a) 11

(c) 13
- (b) 12

(d) 14

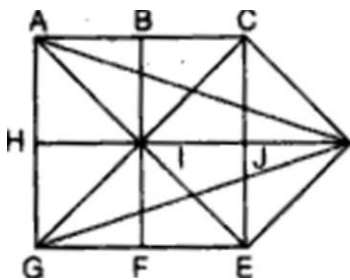
ANSWERS

1. (6) : The figure is labelled as shown :



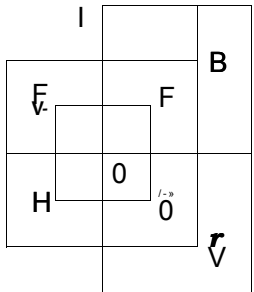
Horizontal lines are DE, FH, IL and BC i.e. 4 in number.
Slanting lines are IM, FN, DO, AC, AB, EM and HN i.e. 7 in number.
∴ Total number of lines is 4 + 7 = 11.

2. (bf) : We can label the figure as shown :



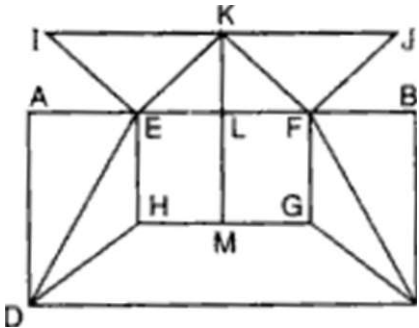
In this figure :
the horizontal lines are AC, HD and GE i.e. 3 in number ;
the vertical lines are AG, BF and CE i.e. 3 in number ;
and the slanting lines are AE, CD, AD, CG, DE and GD i.e. 6 in number.
Thus, there are 3 + 3 + 6 = 12 lines in all.

3. (a) : The figure may be labelled as follows :



Vertical lines are AD, EH, IL, FG, BC and JK i.e. 6.
Horizontal lines are IJ, AB, EF, MN, HG, DC, LK i.e. 7.
∴ Total number of lines is 6 + 7 = 13

4. (f) : The figure may be labelled as follows :



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Triangles

The simplest triangles are UO. JKQ, KLQ. LMQ. MNQ, NOQ, OPQ and PIQ i.e. 8 in number.

The triangles composed of two components are ABQ. BCQ. CDQ, DEO. EFO. FGO. GHO. HAQ, IKQ. KMQ. MOO and OIO i.e. 12 in number.

The triangles composed of four components are ACQ. CEO, EGO, GAQ. IKM. KMO, MOI and OIK i.e. 8 in number.

The triangles composed of eight components are ACE, CEG. EGA and GAC i.e. 4 in number.

Thus, there are $8 + 12 + 8 + 4 = 32$ triangles.

Squares :

Squares composed of two components are IJQP. JKLO. LMNQ and OPON i.e. 4 in number.

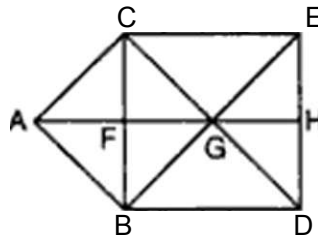
Squares composed of four components are ABOH, BCDO, QDEF and HOFG i.e. 4 in number.

The only square composed of eight components is IKMO.

There is only one square composed of sixteen components which is ACEG.

Hence, there are $4 + 4 + 1 + 1 = 10$ squares in the figure.

8. <c) Wo label the figure as shown :



Count the number of simplest triangles. These are AFC. AFB. BGF. CGF. CGE. BGD. EHG. and DHG. Thus there are 8 such triangles.

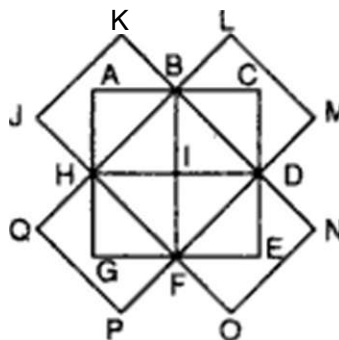
Next, count the number of triangles which are composed of two small triangles each. These are ABC. ACG. CGB. ABG and GDE. Thus, there are 5 such triangles.

Also, count the number of triangles each of which contains three small triangles.

These are BCD. CEB. EDC and EDB. Thus, there are 4 such triangles.

Consequently, there are $8 + 5 + 4 = 17$ triangles in the figure.

9. (0) The figure can be labelled as shown :



The rectangles composed of two components are JKBH, LMDB. NOFD and PQHF i.e. 4 in number.

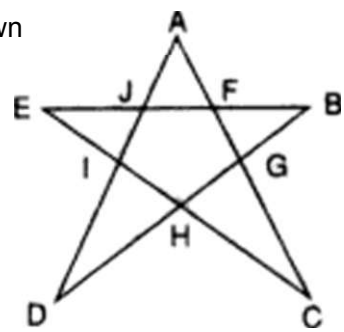
The rectangles composed of four components are ACDH. BCEF. DEGH and FGAB i.e. 4 in number.

The rectangles composed of six components are HLMF. BNOH. POBD and JKDF i.e. 4 in number.

The rectangles composed of eight components are JKNO and PQLM i.e. 2 in number.

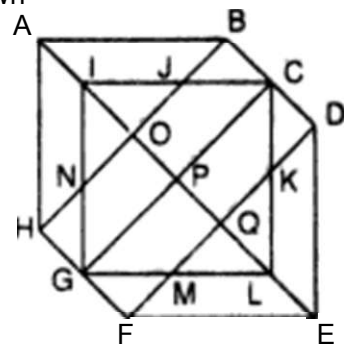
Hence, the total number of rectangles in the figure $\ll 4 + 4 + 4 + 2 \gg 14$

10. (d) The figure is labelled as shown



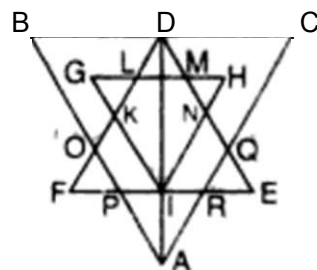
The simplest triangles are AJF. BFG. CGH. OHI and EJI i.e. 5.
The triangles having three parts are AIC. ADG, EHB. EFC and DJB i.e. 5.
There are 10 triangles in the figure

11. (0) The figure is labelled as shown



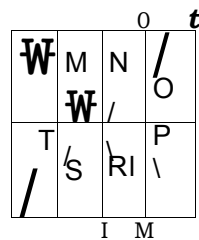
The simplest triangles are BCJ. CDK. KLO. LMO. FGM. GHN. NOI and IJO ie. 8 in number.
The triangles composed of two components each are AOB. DEO. EFO. AOH. GIP CIP. CLP. GLP. KLM and NIJ i.e. 10 in number
The triangles composed of four components each are ABH DEP. ICL. CLG. LGI and GIC i.e 6 in number
∴ Total number of triangles in the figure = 8 • 10 • 6 = 24.

12. (c) The figure may be labelled as shown :



The simplest triangles are GKL. MHN. DLJ. DMJ. ORE. OPF. PIA and IRA i.e. 8.
The triangles having two components each, are BDO. CDO. DLM. PRA. KFI. NEI. HJI. GJI. OKI and DNI i.e. 10
The triangles having four components each, are DIE. DFI. DOA. DQA and DHI i e 5
The triangles having six components each, are DCA and DBA i.e. 2
DEF is the only triangle having eight components
ABC is the only triangle having twelve components
Thus, the'e a re 8*10 + 5 + 2+ i + ?»27 triangles in the figure

13. (b) We label the figure as shown



The simplest squares are BCNM. CDON. HIRQ and SRU i.e. 4.
The squares composed of two components are MNUT. NOPU. UPOR and TURS i.e. 4.
The squares composed of five components are CEFU. GIUF. IKLU and ACUL i.e. 4.
The squares composed of six components are BDPT and TPHJ i.e. 2.
There is only one square i.e. MOQS composed of eight components.
There is only one square i.e. AEGK composed of twenty components.
Hence, there are $4 + 4 + 4 + 2 + 1 + 1 \gg 16$ squares in the figure.

14. (rf) The figure may be labelled as shown :
- B C

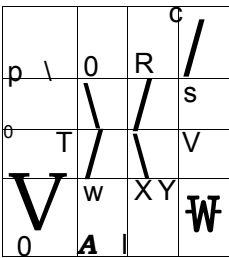
D

H G

F

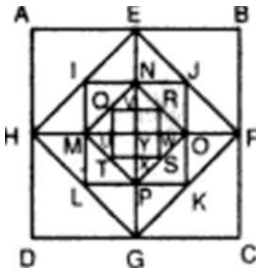
The simplest rectangles are BCJI. UGH. CDEJ and JEFG i.e. 4.
The rectangles composed of two components are BDEI. IEFH. CDFG and BCGH i.e. 4.
The only rectangles composed of four components is BDFH.
Thus, there are $4 + 4 + 1 \gg 9$ rectangles in the figure.

15. (rf) The figure may be labelled as shown :



Now. the simplest triangles are APQ. QTU. UXY. YKC. AEQ. QRU. UVY. YJC. BGS. SRU. UTW. WND, BHS. SVU. UXW and WMD i.e. 16.
The triangles having two components each, are OUS. SUY. WUY and OUY i.e. 4.
The triangles having three components each, are AFU. UIC. AOU. ULC. UOD. ULD, BFU and BIU i.e. 8.
The triangles having four components each, are QSY. SOW. SYW and OWY i.e. 4.
The triangles having six components each, are ABU. ADU. CDU and CBU i.e. 4.
The triangles having seven components each, are ANY. AGY. OMC. QHC. BJW, BEW. SKD and DPS i.e. 8.
The triangles having tweh/e components each are ADC. BDC. ABC and ABD i.e. 4.
Thus, there are in all $16 + 4 + 8 + 4 + 4 + 8 + 4 \ll 48$ triangles in the figure.

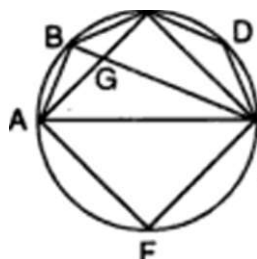
16. (rf) The figure may be labelled as shown :



The simplest squares are VRWY. YWSX. UYXT and OVYU i.e. 4.
The squares composed of four components are QRST. NJOY. OYPK. MYLP and INYM i.e. 5.
The squares composed of seven components are EBFY. YFCG. HYGD and AEYH i.e. 4.

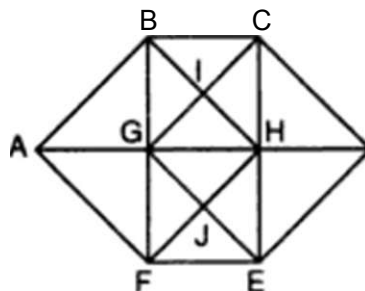
The square MNOP is composed of twelve components.
 The square IJKL is composed of sixteen components.
 The square EFGH is composed of twenty four components.
 The square ABCD is composed of twenty eight components.
 Hence, there are $4 + 5 + 4 + 1 + 1 + 1 + 1 = 17$ squares in the figure

17.(d): We label the figure as shown :



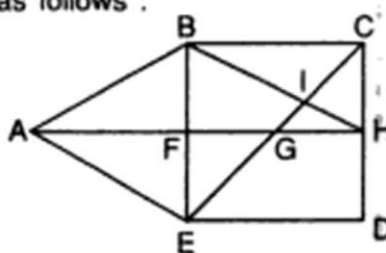
Simplest triangles are ABG, BCG, CDE, GCE, AGE and AFE i.e. 6.
 Triangles composed of two triangles each, are ABC, ABE, ACE and BCE i.e. 4.
 There are $6 + 4 = 10$ triangles in the figure.

18. (d) : The figure is labelled as shown :



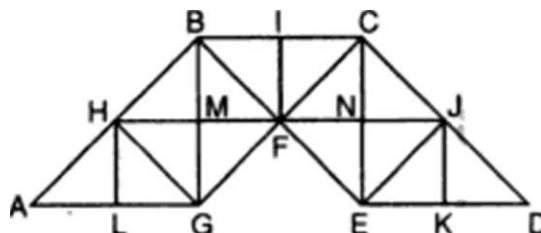
Simplest triangles are ABG, AGF, CHD, HDE, BGI, BCI, HCI, HGI, GHJ, HEJ, EFJ and GFJ i.e. 12.
 The triangles composed of two triangles are ABF, CDE, BCG, BCH, HCG, BHG, GHE, HEF, GFE and GHF i.e. 10.
 The triangles composed of three triangles are AGH, AFH, COG and GDE i.e. 4,
 The triangles composed of four triangles are BHF and CGE i.e. 2.
 $\therefore$ Total number of triangles = $12 + 10 + 4 + 2 = 28$.

19. (d) : The figure may be labelled as follows :



The simplest triangles are CHI, GHI, BCI, EFG, AFE and ABF i.e. 6.
 The triangles composed of two components are ABE, BHF, BEI, CGH, BCH and AEG i.e. 6.
 The triangles composed of three components are ABH, BCE and CED i.e. 3.
 Hence, the total number of triangles in the figure $\Rightarrow 6 + 6 + 3 = 15$.

20. (c) : The figure may be labelled as shown.

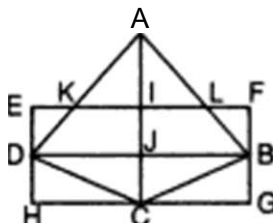


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Simplest triangles are AFE. EFC, CFD. BFD and ABF i.e. 5.
 Triangles having two components are AFC. CFB. ABD and BAE i.e. 4.
 Triangles having three components are ADC and EBC i.e. 2.
 Triangles having five components are ABC i.e. 1.

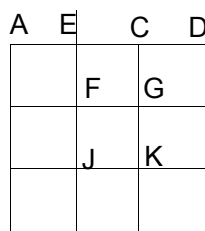
Total number of triangles in the figure - $5 + 4 + 2 + 1 = 12$.

24. (c) : The figure may be labelled as shown.



The simplest triangles are AIK. AIL. EKD. FLB. CDJ. CBJ. CDH and CBG i.e. 8.
 The triangles composed of two components are ADJ. ABJ. AKL and BCD i.e. 4.
 The triangles composed of three components are ADC and ACB i.e. 2.
 The only triangle composed of four components is ADB.
 Thus, there are $8 + 4 + 2 + 1 = 15$ triangles in the figure.

25. (a) : The figure is labelled as shown.



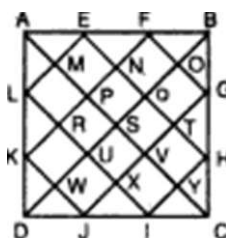
The simplest squares are ABFE. BCGF. CDHG. EFJI. FGKJ. GHLK. IJNM. JKON and KLPO i.e. 9.

The squares composed of four components are ACKI. BDLJ. EFOM and FHPN i.e. 4.

There is only one square i.e. ADPM which is composed of nine components.

Thus, there are $9 + 4 + 1 = 14$ squares in the figure.

26. (c) : The figure is labelled as shown.



Simplest triangles in the figure are AML. LRK. KWD. DWJ. JXI. IYC. CYH. HTG. GOB. BOF. FNE and EMA i.e. 12.

Triangles having two components each, are ALE. KDJ. HIC and BFG i.e. 4.

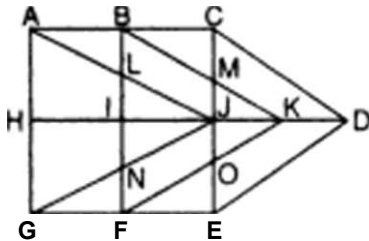
Triangles having three components each, are APK. LUD. DUI. JVC. CVG. HOB. BOE and FPA i.e. 8

Triangles having six components each, are ASD. DSC. BSC. BSA. AFK. LDI. JCG and BEH i.e. 8.

Triangles having twelve components each, are ADC. BDC. ABC and BAD i.e. 4.

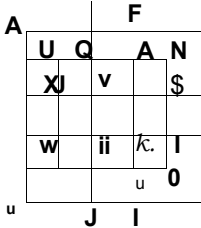
Total number of triangles in the figure - $12 + 4 + 8 + 4 + 4 = 36$.

27. (d) : The figure may be labelled as shown.



Triangles :
Simplest triangles are IU. UN. MJK. OJK. ABL. BCM. GNF and FOE i.e 8.
Triangles composed of two components are AHJ. CJD. LJN. MOK. GHJ and EJD »e 6.
Triangles composed of three components are BIK and FIK i.e. 2.
Triangles composed of four components a'e AGJ ar>d CDE i.e. 2.
The only triangle composed of six components is BFK.
Thus, there are $8 + 6 + 2 + 2 + 1 = 19$ triangles in the figure.
Parallelograms :
Simplest || gms are BUM and FNJO i.e 2.
The || gms. composed of two components are ABIH. HIFG. CBKD and DEFK i.e. 4.
The || gms composed of three components are ABKJ. GFKJ. BCJi and IJEF i.e 4
The only || gm composed of four components is ACFG.
The || gms composed of five components are ACDJ. GEDJ. ACJH and HJEG i.e. 4.
The only || gm composed of six components »s BCEF.
The only || gm composed of ten components is ACEG.
Thus, there are $2 * 4 + 4 + i + 4 + i * i » i 7$ parallelograms in the figure.

28 (d) The figure may be labelled as shown.

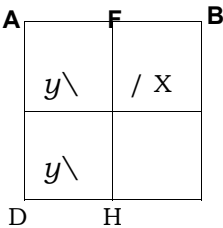


The simplest squares are EFRO. RNSZ. ORZV. MOYX. LXWK. XYB,W YZA,B, ZSTA, SGHT. A,TOU. B,A,UV. WB,VP and VUIJ i.e. 13
The squares having two components each, are AEYL. BFZG. HA,IC and KB,JD i e 4
The squares having four components each, are XZUP. YSOV. ONTB, and MRA,W i.e 4.
The squares having seven components each, are AFA,K. EBHB,. LZID and YGCJ «e 4
The only square having nine components is MNOP.
ABCD is the only square having seventeen components

29. (cfl The figure may be labelled as shown

Simplest triangles are ABL. BCD. DEI⁵. fGP. PGH. HIQ. UO. JKR and KLR i.e. 9..
Triangles composed of two components ace OGS. SGQ. SPI. SRI. KSQ. KSM. FGH, HU and JKL i.e. 9.'
The only triangle composed of four components is KSG.
Triangles composed of five components are CGM, INE. INA and KOC i.e. 4.
Triangles composed of six components are GMK and KOG i.e. 2.
The only triangle composed of ten components is AIE and the only triangle composed of eleven components is CKG.
/. Total number of triangles in the figure «9 + 9 + 1+ 4 + 2 + 1 + 1»27.

30. (a) The figure is labelled as shown below.



Determination of the number of triangles :

Synplest triangles are AIF. IFO. IEO." AIE. FBJ. BJG. JGO. FJO. GKC. HKC. HOK. GOK. OLH. LDH. ELD and ELO i.e. 16.

The triangles having two simple triangles each, are AFE. EDH, HCG. FBG. EOH. HOG. GOF. EOF. AEO. BOG. BOF. AOF. DOE. DOH. GOC and HOC i.e. 16.

The triangles having four simple triangles each, are AOD. DOC. COB. BOA. FEH. EGH. GFH and EFG i.e. 8.

The triangles having eight simple triangles each, are ADC. DBC. ABC and BAD i.e. 4.

The number of triangles in the figure - 16 + 16 + 8 + 4* 44.

Determination of the number of squares :

The squares containing two triangles each, are GJOK. JOIF. IOLE and LOKH i.e 4

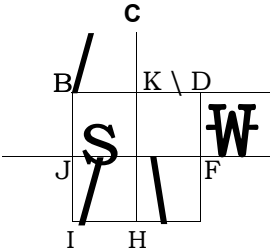
The squares containing four triangles each, are BFOG. AFOE. EOHD and GOHC i.e 4

EFGH is the only square containing eight triangles.

ABCD is the only square containing sixteen triangles.

The total number of squares in the figure - 4 • 4 + 1+ 1 » 10.

31. Tb) The figure is labelled as shown below



Triangles :

Simplest triangles a/e ABJ. BCK. CDK. DEF. BOJ. BOK. KOD. DOF. OFG. HOG. HIO and JOI i.e. 12.

Triangles composed of two components are BCD. ABO. ODE. BOI. BOD. DOG and GOI i.e. 7.

Triangles composed of four components are ACO. COE. DIG. BIG. BID and BDG i.e. 6.

The only triangle composed of eight components is ACE.

Thus, there are 12 + 7 • 6 + 1 * 26 triangles in the grven figure

Squares ::

The squares composed of two components are KDFO. FOHG. JOHI and BKOJ i.e. 4

The only square composed of four components is BCDO.

The only square composed of eight components is BDGI.

Thus, there are 4 + 1 • 1 * 6 squares in the figure.

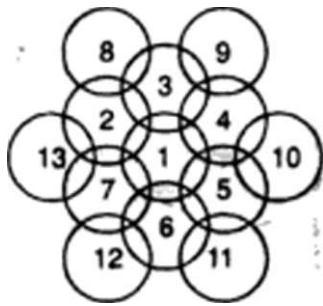
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50. (c) : There are 13 circles in the given figure. This is clear from the foMowing figure in which all the circles have been numbered from 1 to 13.



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APPROACH
TO
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